

MAT 426, Class 25 & 26: The Cauchy-Schwarz Inequality & Fourier Series

Example 1. (*Chebyshev Polynomials*) Let $\mathcal{P}$ be the vector space of all polynomials. Let $w(x) = \frac{1}{\sqrt{1-x^2}}$. Consider the standard basis $\{1, x, x^2, x^3, \dots\}$. Use Gram-Schmidt to construct an orthogonal basis for $\mathcal{P}$ with respect to the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)w(x) dx$ using the normalization that the leading coefficient of $T_n(x)$ is 2^{n-1} .

Definition. Suppose that W is a subspace of a vector space V that has an orthogonal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The **orthogonal projection** $\text{proj}_W(\mathbf{v})$ of a vector $\mathbf{v}$ onto the subspace W is

$$\text{proj}_W(\mathbf{v}) =$$

The **component of $\mathbf{v}$ orthogonal to W** is the vector

$$\text{perp}_W(\mathbf{v}) =$$

CAUCHY-SCHWARZ INEQUALITY Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V . Then,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

with equality holding if and only if $\mathbf{u}$ and $\mathbf{v}$ are scalar multiples of each other.

Proof.

Example 2. Show that the set of functions $\{1, \cos(x), \cos(2x), \cos(3x), \dots, \sin(x), \sin(2x), \sin(3x), \dots\}$ is an orthogonal set with respect to the inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x) dx$

Example 3. (*Fourier Series*) Let $V = \text{Span}\{1, \cos(x), \cos(2x), \cos(3x), \dots, \sin(x), \sin(2x), \sin(3x), \dots\}$ and suppose $f(x) \in V$. Note that each of the functions in V is periodic with period 2π . Therefore, $f(x)$ must be periodic of period 2π ; that is, $f(x + 2\pi) = f(x)$. Also,

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx).$$

Find a formula for the coefficients a_0 , a_k , and b_k .

How do we interpret the Fourier series at the endpoints $[-\pi, \pi]$ and the points of discontinuity?

Example 4. Consider the function $f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 2 & \text{if } x \geq 0 \end{cases}$ Find the Fourier series of $f(x)$.

Example 5. Find the Fourier series of $f(x) = |x|$ on $[-\pi, \pi]$.